MOSAIC

Smart Customer Segmentation & Campaign Administration

Administrator Manual

Flutter + Supabase system for customer segmentation, campaign approval, and role-based access.

Prepared for: Administrators and technical operators

Version: 1.0

Environment: Production / shared Supabase deployment

1. System overview and requirements

Purpose

MOSAIC is a customer segmentation and campaign-management application for analysts, marketing staff, administrators, and customers. It imports customer data, identifies segments, supports approval workflows, and allows customers to view targeted offers and demo orders.

Main components

• Flutter client application for workstation and web access.

• Supabase Auth for sign-in, session restore, and password recovery.

• PostgreSQL tables and row-level security for profiles, customers, segments, campaigns, orders, and audit logs.

• Edge Function for admin-only account invite/delete operations.

• Optional local preview mode for testing without a backend.

Administrator responsibilities

• Create or maintain administrator, analyst, marketing, and customer accounts.

• Assign roles and confirm admin access is granted only by an existing administrator.

• Review pending campaigns, approve or schedule launches, and monitor the audit log.

• Manage segmentation model thresholds (RFM or rule-based) and validate import quality.

• Ensure service keys remain in Supabase secrets and are never embedded in the app.

System requirements

• Flutter SDK 3.12.2+ and a supported browser/device for local testing.

• Supabase project with Auth, Postgres, and Edge Functions enabled.

• Supabase CLI for schema deployment and function publishing.

• Node.js (recommended) for Supabase CLI and local tooling.

• Access to a secure secrets store or environment file for the service role key.

2. Installation and configuration

Install the app

• Clone the project and run: flutter pub get

• Start locally with: flutter run

• For release builds, pass the Supabase settings with --dart-define so client builds do not ship secrets.

Supabase setup

Run the SQL schema from supabase/schema.sql in the Supabase SQL Editor. This creates profiles, customer records, segments, campaign tables, app settings, and row-level security. If signups are created with the wrong role, apply refresh_signup_role_trigger.sql to affect new accounts.

Environment variables

• SUPABASE_URL=https://.supabase.co

• SUPABASE_PUBLISHABLE_KEY=

• SUPABASE_SERVICE_ROLE_KEY= stored only in Supabase Function Secrets

• Do not set the service role in Flutter or pass it as a dart-define.

Deployment commands

supabase link --project-ref supabase db execute --file supabase/schema.sql supabase functions secrets set SUPABASE_SERVICE_ROLE_KEY= supabase functions deploy admin-user

The function verifies admin access using the signed-in user and the profiles table before inviting or deleting users.

3. Administration and maintenance

Account management

• Use the Admin console > Accounts to invite users and set their role to admin, analyst, marketing, or customer.

• Admin access cannot be self-selected during sign-up; assign it after account creation using an existing admin session.

• The database function admin_assign_role(target_user_id, target_role) is the supported role update path.

Data and model management

• Import customer data from CSV or an HTTPS API endpoint; CSV must contain at least name and email.

• Review Data quality findings before committing import results; invalid or duplicate records are flagged for review.

• Run segmentation from the Analyst workspace and review scenario preview before applying threshold changes.

• Manage segmentation settings in Admin > Models; choose RFM or rule-based mode and tune spend/inactivity thresholds.

Campaign and order operations

• Marketing users draft, personalize, and submit campaigns for approval.

• Admins review pending campaigns, approve them, and schedule or launch active campaigns.

• Orders are demo transactions only; Admin can update status from Received to Processing, Fulfilled, or Cancelled.

• Use Audit to inspect staff role changes, model changes, campaign status changes, and customer order events.

Backup and maintenance

• Keep supabase/schema.sql and supabase/demo_seed.sql under version control and document configuration changes.

• Use Supabase backups/scheduled snapshots for production data recovery and export CSV summaries for operational review.

• For local demo mode, data is stored in device storage only and should not be treated as a production database.

4. Troubleshooting and support

Common issues and fixes

• Sign-in fails or user cannot access Admin: verify the account exists in Supabase Auth, the email confirmation policy is acceptable, and the profile role is set correctly.

• Admin invite/delete fails: confirm the Edge Function is deployed, the service role key is set in Supabase secrets, and the requesting user is an administrator.

• Import shows data quality problems: validate CSV headers, ensure name and email are present, and fix invalid email/duplicate rows before importing again.

• Campaign is not visible to customers: confirm campaign status is active, the customer segment matches the target segment, and the customer has Special offers enabled.

• No data appears in app after setup: run supabase/schema.sql, ensure customers were created with a matching email, and verify row-level security policies are active.

Recommended operational checks

• Review the Admin Audit log weekly for role changes, campaign approvals, and model settings updates.

• Verify the deployment after each schema change and re-run the function if account management stops working.

• Rotate the service role key before production use and keep logs of all configuration changes.

• If the app is running in local preview mode, do not rely on local device storage for multi-user or production testing.

Support resources

Use the project README for role-based workflows and the Supabase dashboard for Auth, SQL, logs, and function diagnostics. Keep a copy of the current schema and deployment commands in the project repository for continuity during handover.